Date:

Tugas III 11.09.2024

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Kelas : 1A TNK

A. Diketahui persamaan :

1. $y = 3 \sin x + 4 \cos x$
2. $y = -3 \sin x + 4 \cos x$
3. $y = 3 \sin x - 4 \cos x$

Ubahlah persamaan y tsb bentuk $R \sin (x + \alpha)$ atau $R \sin (x - \alpha)$
Buatkan grafiknya.

1. $3 \sin x + 4 \cos x \equiv R \sin (x + \alpha)$

$3 \sin x + 4 \cos x = R \sin x \cos \alpha + \cos x \sin \alpha$

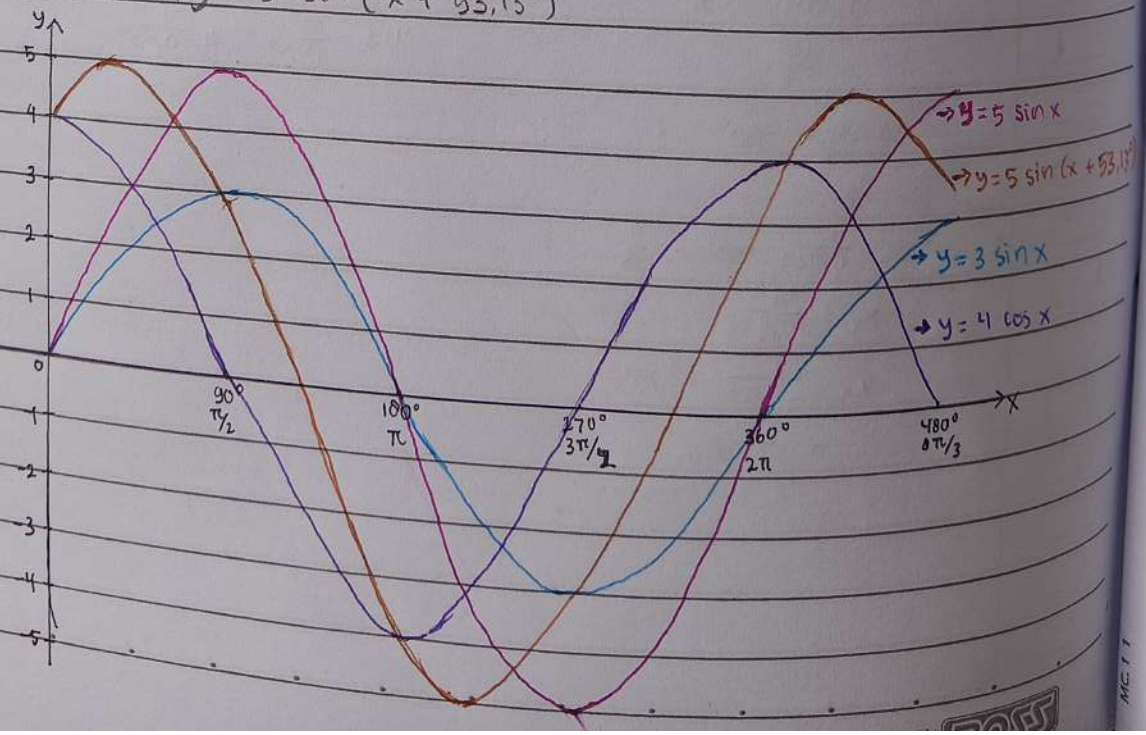
$a = 3 \quad b = 4$

$R = \sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\alpha = \text{arc tan} \left(\frac{b}{a}\right)$

$= \text{arc tan} \left(\frac{4}{3}\right) = 53{,}13°$

Jadi, $y = 5 \sin (x + 53{,}13°)$



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2. $-3\sin x + 4\cos x \equiv R\sin(x+\alpha)$

$-3\sin x + 4\cos x = R\sin x\cos\alpha + \cos x\sin\alpha$

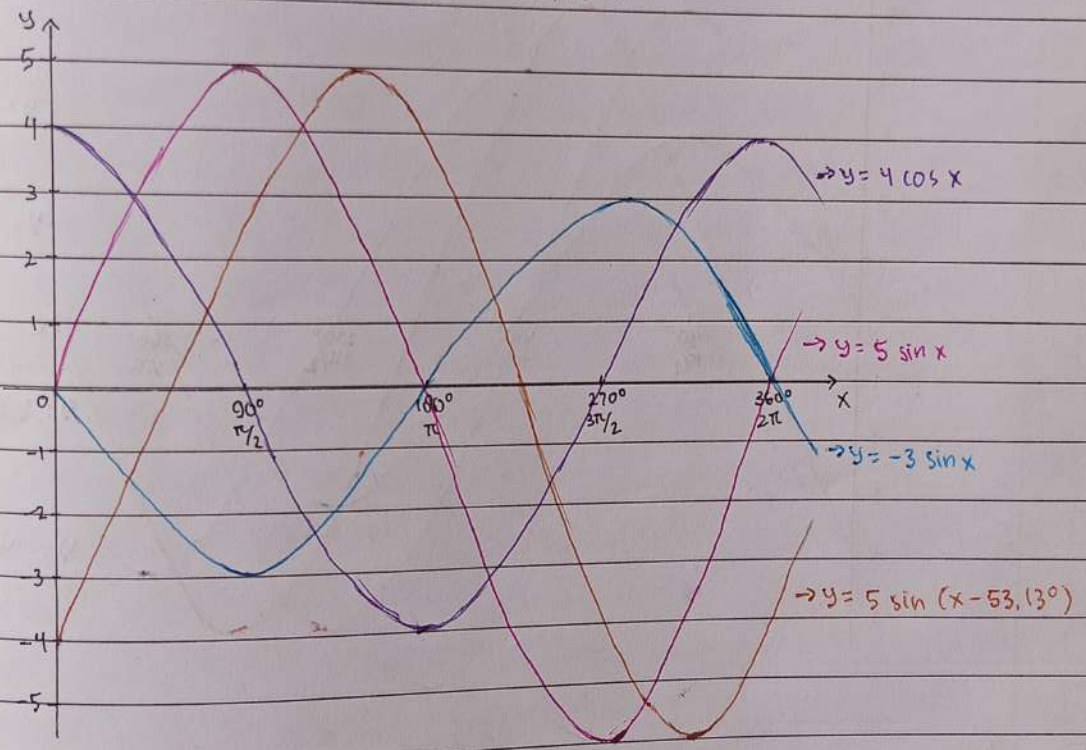
$a = -3 \quad b = 4$

$R = \sqrt{a^2+b^2} = \sqrt{(-3)^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5$

$\alpha = \arctan\left(\frac{b}{a}\right)$

$= \arctan\left(-\frac{4}{3}\right) = -53{,}13°$

Jadi, $y = 5\sin(x - 53{,}13°)$



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3. $3 \sin x - 4 \cos x \equiv R \sin (x - \alpha)$

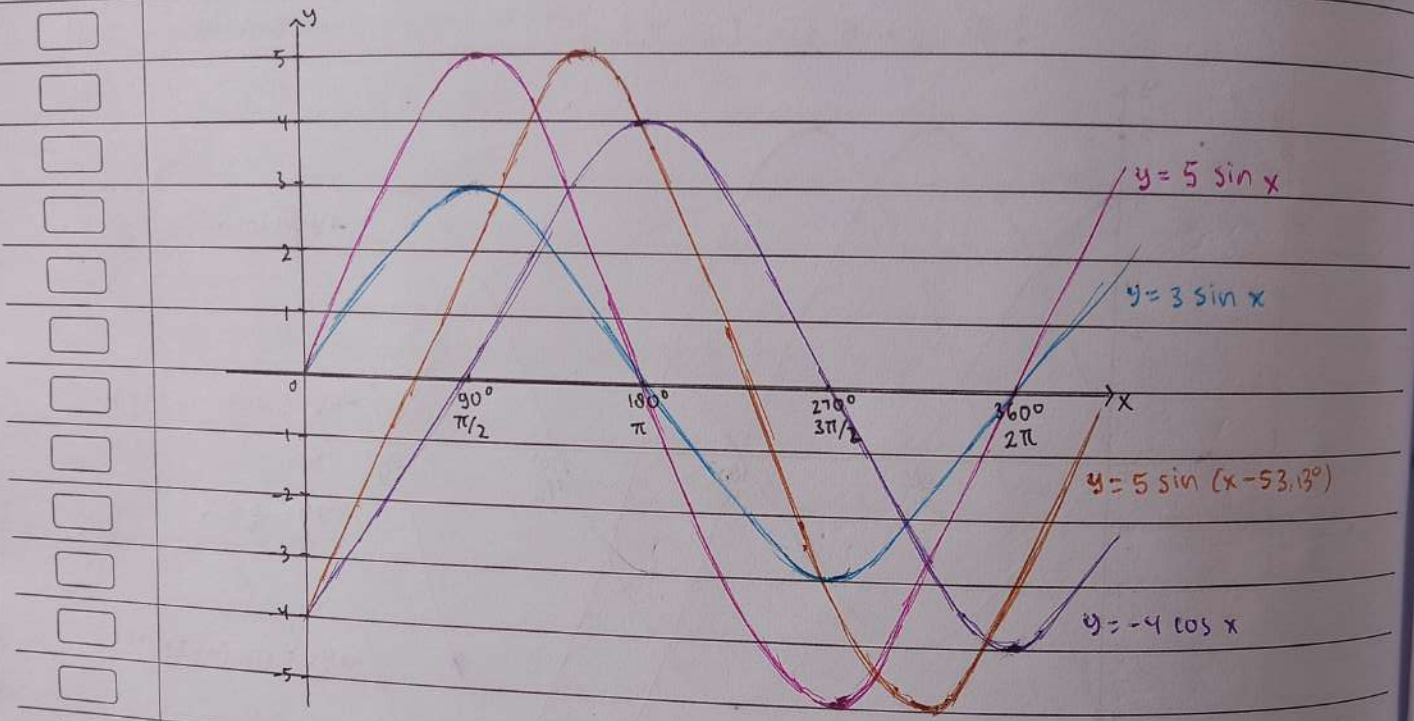
$3 \sin x - 4 \cos x = R \sin x \cos \alpha - \cos x \sin \alpha$

$a = 3 \quad b = -4$

$R = \sqrt{a^2 + b^2} = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\alpha = \arctan \left(\frac{b}{a}\right)$

$= \arctan \left(-\frac{4}{3}\right) = -53{,}13°$ Jadi, $y = 5 \sin (x - 53{,}13°)$



B. Selesaikan bagian A ke bentuk $R \cos (x + \alpha)$ atau $R \cos (x - \alpha)$.

1. $y = 3 \sin x + 4 \cos x$

$4 \cos x + 3 \sin x \equiv R \cos (x + \alpha)$

$4 \cos x + 3 \sin x = R \cos x \sin \alpha + \sin x \cos \alpha$

$a = 4 \quad b = 3$

$R = \sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

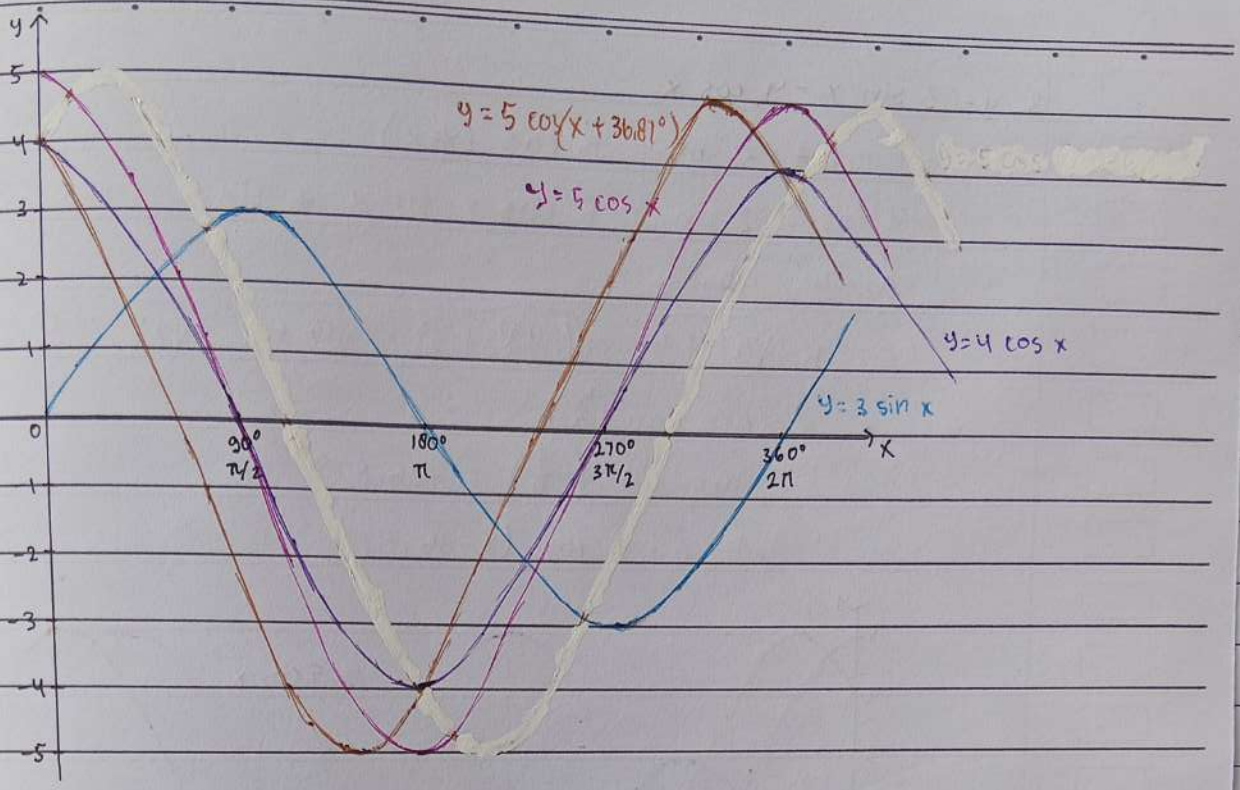
$\alpha = \arctan \left(\frac{b}{a}\right)$

$= \arctan \left(\frac{3}{4}\right) = 36{,}87°$

Jadi, $y = 5 \cos (x + 36{,}87°)$

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2. $y = -3 \sin x + 4 \cos x$

$4 \cos - 3 \sin x \equiv R \cos(x - \alpha)$

$4 \cos - 3 \sin x = R \cos x \sin \alpha - \sin x \cos \alpha$

$a = 4 \qquad b = -3$

$R = \sqrt{a^2 + b^2} = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$

$\alpha = \arctan\left(\frac{b}{a}\right)$

$= \arctan\left(-\frac{3}{4}\right) = -36,87°$

Jadi, $y = 5 \cos(x - 36,87°)$



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3. $y = 3\sin x - 4\cos x$

$-4\cos + 3\sin \equiv R\cos(x+\alpha)$

$-4\cos + 3\sin = R\cos x \sin\alpha + \sin x \cos\alpha$

$a = -4 \quad b = 3$

$R = \sqrt{a^2 + b^2} = \sqrt{(-4)^2 + 3^2} = \sqrt{16+9} = \sqrt{25} = 5$

$\alpha = \arctan\left(\frac{b}{a}\right)$

$= \arctan\left(-\frac{3}{4}\right) = -36{,}87°$

Jadi, $y = 5\cos(x - 36{,}87°)$

